DIAGNOSIS OF OLIVE KNOT DISEASE STAGES THROUGH IMAGE PROCESSING AND BACILLUS SUBTILIS TREATMENT

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TÜBİTAK

PROJECT ABSTRACT

Olives are important to Turkey's economy and food supply. Identifying agricultural factors that reduce yield and harvest quality is therefore essential. Olive knot disease is one of the major problems affecting olive trees and fruit quality. This project investigated early identification of disease stages using artificial intelligence and possible treatment. Twelve saplings were divided into four groups of three. Measurements were recorded at intervals for four months, beginning while all saplings were healthy. Pathogenicity and oxidase tests, IAA purification, and analysis were performed. Compared with the control group, the saplings receiving B. subtilis showed faster leaf development, and its application appeared to slow adverse effects.

All analyses were transferred to a CNN-based image-processing model trained with Keras. Collected photographs of disease symptoms were organized into four stages and loaded through ImageDataGenerator. Training results were examined for overtraining, and the correct-detection rate was calculated and plotted. By distinguishing four disease stages, the model was intended to help farmers sort produce by quality and avoid disadvantaging consumers. The project proposed that its AI system could provide rapid and definitive olive-knot diagnosis and that positive treatment results could protect Turkey's position in the olive market.

PROJECT AIM

Olives matter to Turkey's economy and nutrition. Olive knot, caused by Pseudomonas savastanoi pv. savastanoi (Pss), affects olive yields and quality. The poster proposes alternative approaches to bacterial disease control where chemical or antibiotic use is constrained. Its aim was to reduce the expertise and effort required of farmers to identify and stage disease, support produce classification, and investigate B. subtilis as a treatment. An image-processing AI model was developed for the diagnosis and stage classification task.

1. INTRODUCTION

1.1 OLIVE PRODUCTION AND OLIVE KNOT IN TURKEY

Olive trees are native to Mediterranean climates, including regions of Turkey (Gemas et al., 2004). Their fruit is used for table olives and oil, while leaves have pharmaceutical uses (Özkaya et al., 2010). Olives are among the important products in Turkey's agricultural economy and a source of income for growers (Akay, 1996). Figure 1 shows olive-growing areas. The number of olive trees reportedly increased about 2.5-fold over the preceding twenty years, although production did not rise at the same rate in recent years (TÜİK, 2015).

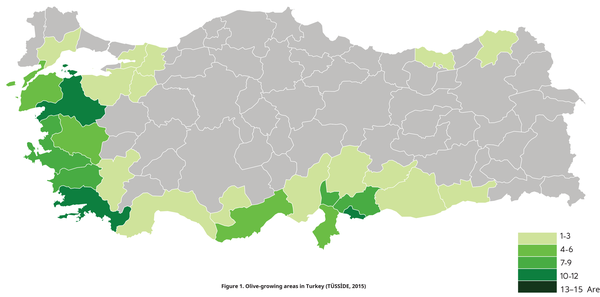


Figure 1. Olive-growing areas in Turkey (TÜSSİDİ, 2015)

	2015	2016	2017	2018	2019	Change(%)
Total olive-growing area (thousand da)	8.369	8.455	8.461	8.644	8.792	1,7
Total olive trees (thousand)	171.992	173.758	174.594	177.844	182.076	2,4
Total olive production	1.700.000	1.730.0000	2.100.000	1.500.0000	1.525.000	1,7
Table-olive production	400.000	430.000	460.000	426.995	415.000	-3,0
Oil-olive production	1.300.0000	1.300.0000	1.640.000	1.073.472	1.110.000	3,4
Olive-oil production	185.000	195.000	287.041	193.613	224.595	16,0
Olive-oil exports³	12.831	17.819	50.217	65.940	43.862	-33,5
Olive-oil import⁴	2.632	1.307	16	4.844	23.401	383,1

Source: TÜİK, UZZK (accessed January 2021). 1/ Change over the last two years. 2/ Provisional data (UZZK). 3/ Customs code 150910.

Figure 2. Turkey's olive and olive-oil data (tonnes)

1.2 WHAT IS BACILLUS SUBTILIS?

Bacillus bacteria are an important part of soil microflora and belong to the Bacillaceae family. They are Gram-positive, aerobic or facultatively anaerobic rods. Bacillus species can produce bacteriocins and enzymes. Some inhibitory compounds can suppress or kill food pathogens and spoilage organisms, including antagonistic bacteriocins. These compounds can give the bacterium a competitive advantage, while it is not affected by its own bacteriocin (Temiz, 1998). Agricultural applications of B. subtilis have shown positive effects relative to controls.

1.3 WHAT IS IMAGE PROCESSING?

Image processing applies techniques to digital images to extract useful information or improve their quality. For example, scanning photographs or documents can reduce quality; processing can limit the visible degradation. Satellite images used in mapping services can also be enhanced with image-processing techniques.

2. MATERIALS AND METHODS

2.1 MATERIALS

- Ethanol, methanol, acetone
- Dichloromethane
- tert-Butyl methyl ether
- Trimethylamine
- Sodium carbonate
- Sodium hydroxide
- Sodium potassium tartrate; sodium dihydrogen phosphate
- Copper sulfate
- Potassium dichromate
- 1,5-Diphenylcarbazide
- Citric acid
- Hydrochloric acid
- Phosphoric acid
- Folin–Ciocalteu reagent
- FeCl3, FeCl2, NH4SCN, Na2HPO4, K3Fe(CN)6, TCA (trichloroacetic acid)
- 5000 mL Sim Bacil (Bacillus subtilis)
- Whatman No. 1

2.2 METHOD

2.2.1 Study population:

Twelve Edremit olive saplings were planted in 15 L black pots with soil and fertilizer mix. B. subtilis (Simbiyotek Biological Products) was applied to study its effects on healthy and infected saplings. The plants were divided into four groups. Their root zones were held for five minutes in the solutions at the doses shown in Table 1. After planting and watering, the root-zone treatment was repeated every 30 days for four months.

Group 1. Healthy

Group 2. Pseudomonas savastanoi

Group 3. Bacillus subtilis (4%)

Group 4. Bacillus subtilis + Pseudomonas savastanoi

Groups	Health before experiment Intervention on plant	Dose
Control (K)	Tap water	0,5 L
Group 2 (G2)	B. subtilis application	4% solution
Group 3 (G3)	P. savastanoi (Pss) inoculation	Droplets containing 10-6 CFU Pss per mL (Ustun, 2021)
Group 4 (G4)	B. subtilis + P. savastanoi application	4% solution

Table 1. Control groups and applied doses

2.2.2 Study plan:

The three-phase plan began with morphological observation, followed by in vitro and in vivo laboratory work to assess the interventions' effects on saplings. IAA and Iy-IAA levels and ratios were examined for their potential use in early Pss detection. Plant height, stem diameter, shoot length, and leaf count were recorded monthly over four months. At the end, healthy and symptomatic branches were sampled. Samples showing typical olive-knot galls were wrapped, bagged, labeled, and brought to the laboratory on ice; isolation was performed within 24–48 hours.



Figures 3–4. HPLC device and data processing

Sim Bacil was used as the source of Bacillus subtilis.

Sim Bacil composition: total organic matter 10%; zinc 3%; protease 200 U/g; amino acids including aspartic acid, glutamic acid, asparagine, serine, histidine, glycine, threonine, citrulline, arginine, alanine, tyrosine, cystine, valine, methionine, tryptophan, phenylalanine, isoleucine, leucine, lysine, and proline. Total live microorganisms: B. subtilis 1×10⁹ CFU/g. The natural isolate B. subtilis KUEN 1581 is registered in the Istanbul University Medical Faculty microorganism culture collection on behalf of Simbiyotek A.Ş.

In the bacterial and plant experiment, Pseudomonas savastanoi pv. savastanoi was isolated, and IAA purification and oxidase tests were performed on the samples. The reference IAA standard curve was compared with HPLC results, bacterial concentration was determined, and the experimental progression was recorded. The poster reports pathogenicity confirmation.

Stages performed: pathogenicity test and recovery of re-isolates; isolation of P. savastanoi pv. savastanoi; oxidase test; IAA purification and analysis.



Figure 5. Cross-sections of P. savastanoi (Moreno-Pérez et al., 2020)

Values

Column: µBondapak C18 (Waters; 3.9 × 300 mm)

Detector: DAD

Wavelength: 254 nm for ABA and GA3; 280 nm for IAA; 270 nm for Z

Mobile-phase flow: 1 mL/min

Column-oven temperature: 40 °C

Injection volume: 20 µL

Analysis time: 15 min + 5 min column cleaning (post-run)

Gradient mobile-phase flow:

Mobile phase A: HPLC-grade acetonitrile with 0.05% acetic acid

Mobile phase B: HPLC-grade water

IAA extract obtained:

Read by HPLC at 280 nm; IAA calculated from the prepared standard curve.

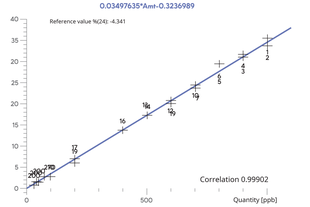


Chart 1. Standard IAA curve used for quantitative HPLC measurement of IAA in olive leaf, knot, and shoot samples at 280 nm

2.2.7 Building the CNN image-processing model:

The planned AI evaluation compared stage-classification outputs with observation data and proposed checking the stage predictions against an external test set unavailable to the model during training.

An AI model was trained to help farmers classify produce and distinguish plant conditions if olive knot spread in a field, supporting decisions about produce grouping and potential treatment.

Classifying photographs of olive plants showing P. savastanoi infection at different stages is a supervised machine-learning classification problem.

The training dataset used four stages based on published staging and technical instructions from Turkey's General Directorate of Agricultural Research and Policies (TAGEM). Plants were staged and photographed. The photographs were placed in a folder hierarchy compatible with TensorFlow Keras DataGenerator, shown below.

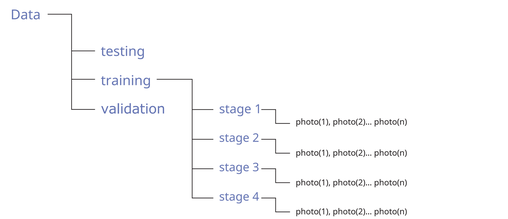


Figure 6. File hierarchy (folders in blue, individual files in black)

With the indicated folder hierarchy, Keras DataGenerator groups classification outputs as in Figure 6 and reduces model-training preparation time.

The training images showed shared characteristics and distinguishable differences, including knots of different sizes against olive-plant backgrounds. Early prototypes tried transfer learning with pretrained models such as VGG16 or TransferNet. After inspecting the dataset, a custom convolutional neural network (CNN) was selected. Keras checkpoints were used, and the model's weight output was inspected in Netron (Roeder, 2022).

Hyperparameters, including activation function and other model settings, were chosen for higher prediction accuracy while considering loss and the risk of overfitting.

The final trained model reported a stage-classification value of 0.88333 (about 88% accuracy) and was saved in .h5 format. The software was written in Python in JupyterLab.

```
# -*- coding: utf-8 -*-
"""ErenIA A

Automatically generated by Colaboratory.

Original file is located at
    https://colab.research.google.com/drive/1jVIKaLvfzyQZ1safxrWFRqRfDcze41Kj
"""

!wget -O data.zip https://ink.ci/TFK7s

#Sıkıştırılmış verinin genişletilme işlemi
!unzip data.zip -d "data/"

!ls data

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import datetutil
import os
from tqdm import tqdm
from keras.activations import tanh
from keras.activations import sigmoid
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from keras.models import Sequential
from keras.layers import Dense
from keras.layers import LSTM
from keras.layers import Dropout
from keras.layers import Activation
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.random import set_random_seed
from tensorflow.keras.layers import Flatten, Dense, Dropout
from tensorflow.keras.regularizers import l1, l2, l1l2
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint

from keras.models import Sequential
from keras import optimizers, losses
from keras.layers import Conv2D, Flatten, MaxPooling2D, Dense, Dropout, GlobalAveragePooling2D

valdata = 'data/valdation'
testdata = 'data/testing'
traindata = 'data/training'
datadir = 'data/'
datagen = ImageDataGenerator()

"""## AUGMENTATION (Veri Üzerinde Başkalaşım İşlemi)

"""

set_random_seed(0)
np.random.seed(0)

train_datagen = ImageDataGenerator(
        rescale = 1./255,
        rotation_range = 20,
        width_shift_range = 0.2,
        height_shift_range = 0.2,
        horizontal_flip = True,
        vertical_flip = True,
        fill_mode='nearest'
)
validation_datagen = ImageDataGenerator(
        rescale = 1./255
)
test_datagen = ImageDataGenerator(
        rescale = 1./255
)

path = 'data'

img_shape = (224, 224, 3)

train_batch_size = 156
val_batch_size = 12

train_generator = train_datagen.flow_from_directory(
            path + '/training',
            target_size = (img_shape[0], img_shape[1]),
            batch_size = train_batch_size,
            class_mode = 'categorical',)

validation_generator = validation_datagen.flow_from_directory(
            path + '/validation',
            target_size = (img_shape[0], img_shape[1]),
            batch_size = val_batch_size,
            class_mode = 'categorical',
            shuffle=False)

test_generator = test_datagen.flow_from_directory(
            path + '/testing',
            target_size = (img_shape[0], img_shape[1]),
            batch_size = val_batch_size,
            class_mode = 'categorical',
            shuffle=False,)

model = Sequential()

model.add(tf.keras.layers.Conv2D(64, (5, 5), padding='same', activation='relu'))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2, 2), strides=(2,2)))
model.add(tf.keras.layers.Dropout(0.15))
model.add(tf.keras.layers.BatchNormalization(input_shape=x_train.shape[1:]))
model.add(tf.keras.layers.Conv2D(128, (5, 5), padding='same', activation='relu'))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2, 2)))
model.add(tf.keras.layers.Dropout(0.15))
model.add(tf.keras.layers.Flatten())
model.add(tf.keras.layers.Dense(1024))
model.add(tf.keras.layers.Activation('relu'))
model.add(tf.keras.layers.Dropout(0.5))
model.add(tf.keras.layers.Dense(10))
model.add(tf.keras.layers.Activation('softmax'))

from google.colab import drive
drive.mount('/content/gdrive')

#lr = öğrenme hızı (Learning Rate)
model.load_weights('/content/gdrive/My Drive/eth.h5')
model.compile(loss='categorical_crossentropy',
              optimizer=adam(lr=1e-4),
              metrics=['acc'])

es = EarlyStopping(monitor='val_loss', mode='min', verbose=1, patience=10)
mc = ModelCheckpoint('ZeytinTespit.h5', monitor='val_acc', mode='max', verbose=1, save_best_only=True)

history = model.fit_generator(
    train_generator,
    steps_per_epoch=train_generator.samples/train_generator.batch_size ,
    epochs=25,
    validation_data=validation_generator,
    validation_steps=validation_generator.samples/validation_generator.batch_size,
    verbose=0,
    callbacks = [es, mc],)

train_acc = history.history['acc']
val_acc = history.history['val_acc']
train_loss = history.history['loss']
val_loss = history.history['val_loss']

epochs = range(1, len(train_acc) + 1)

plt.plot(epochs, train_acc, 'b*-', label = 'Training accuracy')
plt.plot(epochs, val_acc, 'r', label = 'Validation accuracy')
plt.title('Training and validation accuracy')
plt.legend()

plt.figure()

plt.plot(epochs, train_loss, 'b*-', label = 'Training loss')
plt.plot(epochs, val_loss, 'r', label = 'Validation loss')
plt.title('Training and validation loss')
plt.legend()

model.predict

predictions = model.predict_generator(generator=validation_generator)
y_pred = [np.argmax(probas) for probas in predictions]
y_test = validation_generator.classes
class_names = validation_generator.class_indices.keys()

from keras.preprocessing import image

img_path = 'data/testing/paper/paper258.jpg'

img = image.load_img(img_path, target_size=(224, 224, 3))
img = image.img_to_array(img, dtype=np.uint8)
img=np.array(img)/255.0

plt.title("Loaded Image")
plt.axis('off')
plt.imshow(img.squeeze())

p=model.predict(img[np.newaxis, ...])

#print("Predicted shape",p.shape)
print("Maximum Probability: "+np.max(p[0], axis=-1))
predicted_class = labels[np.argmax(p[0], axis=-1)]
print("Classified:"+predicted_class)
```

Code shown is a 2022 development representation, not the finished implementation.

4. FINDINGS

4.1 Morphological and phenological observations

To evaluate the interventions, leaf count, stem diameter (mm), and shoot length (cm) were recorded weekly. At the end of the trial, leaf and root counts, root and shoot length, stem diameter, and root weight were measured and recorded for rooted cuttings.

Parameters	Before experiment	Month 1	Month 2	Month 3	Month 4
Groups	K | G2 | G3 | G4	K | G2 | G3 | G4	K | G2 | G3 | G4	K | G2 | G3 | G4	K | G2 | G3 | G4
Gall count	0 | 0 | 0 | 0	0 | 0 | 0 | 0	0 | 0 | 3 | 2	0 | 0 | 4 | 2	0 | 0 | 6 | 3
Leaf count	187 | 174 | 191 | 183	196 | 187 | 191 | 186	202 | 199 | 185 | 189	210 | 214 | 177 | 184	215 | 222 | 170 | 178
Active stem ratio (A/T)	1 | 1 | 1 | 1	1 | 1 | 0.98 | 1	1 | 1 | 0.92 | 0.96	1 | 1 | 0.83 | 0.90	1 | 1 | 0.74 | 0.87
Pathogenicity	- | - | - | -	- | - | + | +	- | - | + | +	- | - | + | +	- | - | + | +
IAA (µg)	60 | 63 | 61 | 66	59 | 62 | 73 | 67	61 | 60 | 90 | 67	61 | 60 | 106 | 73	60 | 61 | 123 | 75
Gram test	- | - | - | -	- | - | + | +	- | - | + | +	- | - | + | +	- | - | + | +
Oxidase test	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -	- | - | - | -
Stage result	0 | 0 | 0 | 0	0 | 0 | 0 | 0	0 | 0 | 1 | 1	0 | 0 | 3 | 2	0 | 0 | 4 | 3

Table 2. Longitudinal measurements

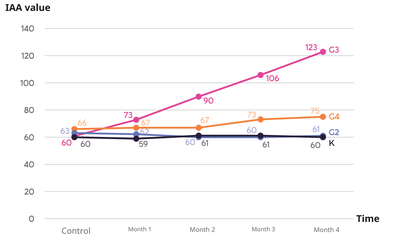


Chart 2 below shows the change in IAA values over time based on the findings table.

Chart 2. IAA values by experimental group

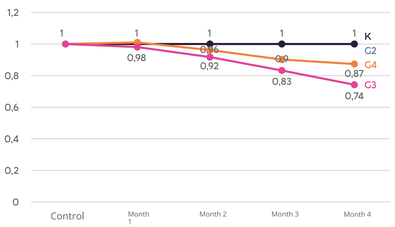


Figure (Chart 3) shows IAA values for control (K), G2, G3, and G4. The highest value was in G3, the group with the highest infection level, followed by G4 and the healthy K and G2 groups.

Chart 3. Active stem ratios

5. RESULTS AND DISCUSSION

The interventions in the experiment table were applied at the start. By the end of month two, galls were observed in the Pss-inoculated G3 and G4 groups, confirming infection. Values from every experimental stage were recorded; quantitative and qualitative differences were measured, and variances were tabulated. Changes in disease stages were identified using the image-processing model deployed to a phone app in TFLite format. The error, accuracy, and correlation results remained stable from training, which the poster treated as confirmation of model effectiveness. Considering the recorded stage changes and measurements, the stage transition (delta) in the B. subtilis plus Pss group (G4) was slower than in G3. Active stem ratios in G3 and G4 were lower than those in the control and B. subtilis-only group (G2). B. subtilis application was found to slow disease-stage development under Pss infection; active stem area, necrosis, and plant-survival observations in G4 were more favorable than in Pss-only plants.

Slower stage transitions and AI-based distinction between stages were considered potentially useful for produce classification and economic outcomes.

6. RECOMMENDATIONS

The model appears promising for early identification and staging, using the project's morphological, hormonal, and biochemical observations. Larger studies are needed to confirm preventive or therapeutic effects of B. subtilis. More photographs of staged Pss galls from different fields could increase training data and may improve classification accuracy.

Future work would use earlier images of each B. subtilis-treated plant to estimate plant-specific infection progression and its rate relative to a normal course.

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